



Does conditioning leave a structural memory?

Large-area evidence for the dislocation picture

Yinon Ashkenazy

Racah Institute of Physics, The Hebrew University of Jerusalem

I. Popov (HUJI Nano) · V. M. Bjelland, W. Wuensch, L. Millar, S. Calatroni (CERN)



2026-10-02

Does conditioning leave a structural memory?

This talk is the sequel to last year's MeVArc talk. In 2025 I showed our dislocation model of conditioning and cross-sectional STEM that found a modified near-surface layer, and I closed by asking for more comparative forensics—other materials, more samples. This year answers part of that: a large-area copper measurement that tests the model's central prediction, and a stainless-steel experiment that will test whether the effect generalizes. E_S is deliberately kept out of the introduction; it appears once, near the end.

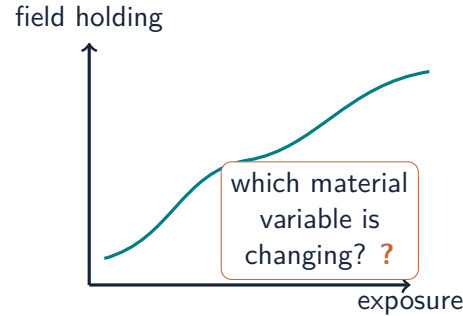


Conditioning works. What does it change?



- ▶ Conditioning is how every high-gradient device reaches its usable field.
- ▶ Breakdown rate has an extreme field dependence, $\sim E^{30}$.
- ▶ It tracks the number of **high-field pulses**, not the number of breakdowns.
- ▶ The field itself drives conditioning, not the arcs.

Something in the metal stores the improvement. We do not know what.



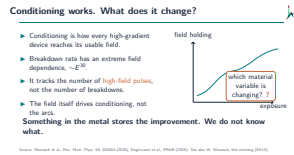
Source: Wuensch et al., Rev. Mod. Phys. 98, 025004 (2026); Degiovanni et al., PRAB (2016). See also W. Wuensch, this morning (09:15).

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Does conditioning leave a structural memory?

└ Conditioning works. What does it change?

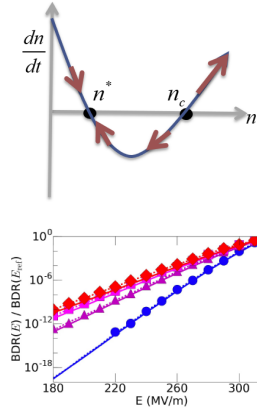
Open on the operational fact and the puzzle. Do not introduce E_S here. Two macroscopic clues already constrain the answer: the near-thirtieth-power field dependence, and that conditioning follows pulse count rather than breakdown count. Both say the applied field, not the arc, is doing the work—so we should look for a field-driven change in the metal itself.



Our model: conditioning as collective dislocation dynamics



- Field holding rises FCC→BCC→HCP — the order of barriers to dislocation motion.
- The **MDDF** model: breakdown nucleates when the mobile-dislocation population, driven by Maxwell stress, runs away.
- One model gets the $\sim E^{30}$ rate, its temperature dependence, and the dark-current spikes.



Central prediction: conditioning *evolves the near-surface dislocation structure*.

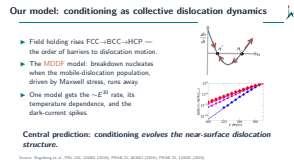
Source: Engelberg et al., PRL 120, 124801 (2018); PRAB 22, 083501 (2019); PRAB 23, 123501 (2020).

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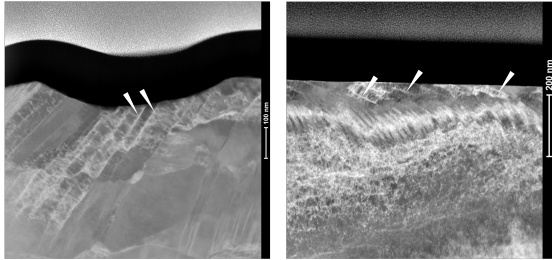
Does conditioning leave a structural memory?

└ Our model: conditioning as collective dislocation dynamics

The MDDF model is our own (2018–2020). The cartoon shows the birth–death dynamics of the mobile-dislocation population: below a critical density the population decays, above it a runaway triggers nucleation. The plot is the reproduced field dependence of the breakdown rate. The key line for this talk is the prediction highlighted at the bottom: conditioning should leave an evolving dislocation structure just below the surface. Everything after this slide is a test of that sentence.



STEM already saw it, at single spots



Cross-sectional STEM of conditioned hard (as-machined)

Cu: dislocation-poor zone in the top ~200 nm.

Seen at one spot. The large-area comparison was still missing.

Source: Jacewicz et al., JAP 137, 193302 (2025); HUJI STEM (I. Popov), MeVArc 2025; Korsbäck et al., PRAB 23, 033102 (2020).

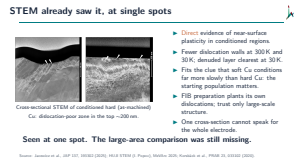
- ▶ **Direct** evidence of near-surface plasticity in conditioned regions.
- ▶ Fewer dislocation walls at 300 K and 30 K; denuded layer clearest at 30 K.
- ▶ Fits the clue that soft Cu conditions far more slowly than hard Cu: the starting population matters.
- ▶ FIB preparation plants its own dislocations; trust only large-scale structure.
- ▶ One cross-section cannot speak for the whole electrode.

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Does conditioning leave a structural memory?

└─ STEM already saw it, at single spots

These are our 2025 STEM images. The dark band is the protective coating over the surface; just beneath it the field-exposed region is nearly free of dislocation walls, while the reference keeps its walls continuous up to the surface. The published wording is conservative: at 300 K conditioning significantly reduces the number of dislocation walls; the denuded zone is called clearly visible for the 30 K cathode. The paper gives no thickness; the 200 nanometers is read from the images and is the number the manuscript uses. The hard-versus-soft copper contrast (Korsbäck 2020) is an independent macroscopic clue pointing the same way: hard copper conditioned within about 25 million pulses in that test, while one soft sample only began to saturate near 600 million and the others stayed unsaturated longer, consistent with different starting dislocation populations. Two lessons carried forward: FIB sample preparation itself generates dislocations and loops—real red herrings—so only larger-scale structure is trustworthy; and TEM/STEM is intrinsically local. Last year I closed by asking for more comparative forensics: other materials, more samples. That is exactly the gap the next slides fill.



A tiny stress, repeated a billion times

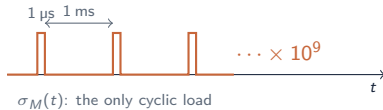


Maxwell stress

$$\sigma_M = \frac{\epsilon_0 E^2}{2}$$

at 80 MV/m: **0.028 MPa**

$\sim 10^3 \times$ below the yield of annealed Cu



The model predicts a large-area signature. Last year we asked for it; this year we looked.

- ▶ Fatigue at 10^8 – 10^{10} cycles rearranges dislocations irreversibly, well below yield.
- ▶ Here: $\sim 10^9$ conditioning pulses.
- ▶ Pulsed DC means **no RF pulsed heating** and no bulk currents. Maxwell stress is the only cyclic load.
- ▶ MDDF is **threshold-free**: no minimum stress amplitude.

Source: Mughrabi, MMTB 40, 431 (2009); Stanzl-Tschegg et al., Int. J. Fatigue 29, 2050 (2007). Field-driven deformation at the nanoscale:

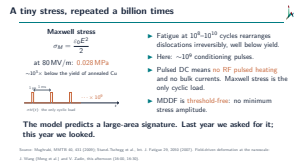
J. Wang (Meng et al.) and V. Zadin, this afternoon (16:00, 16:30).

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Does conditioning leave a structural memory?

└ A tiny stress, repeated a billion times

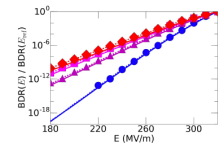
Head off the yield-stress objection: the Maxwell stress is three orders of magnitude below yield. The point is not amplitude but accumulation. VHCF shows that oscillatory loading well below yield still rearranges dislocations irreversibly when sustained for a billion cycles, which is our regime. The pulsed-DC system is also a cleaner experiment than RF structures: there is no RF magnetic field, no pulsed surface heating, and no bulk Ohmic current, so the Maxwell stress is the only cyclic mechanical load—earlier RF-based studies could never fully separate thermal fatigue from field effects. And the MDDF description is threshold-free, so it does not need the stress to exceed any yield-like value. That argument is the bridge from the 2025 model-and-microscopy story to this year's large-area measurement.



Last year we asked for more forensics. Here it is.

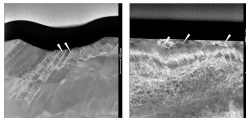


MeVArc 2025



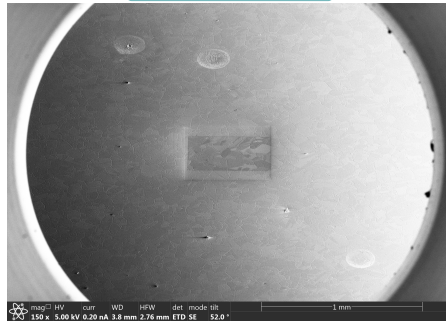
MDDF model

Closed with: other materials? more samples?



local STEM denuded zone

MeVArc 2026



Large-area EBSD across one cathode

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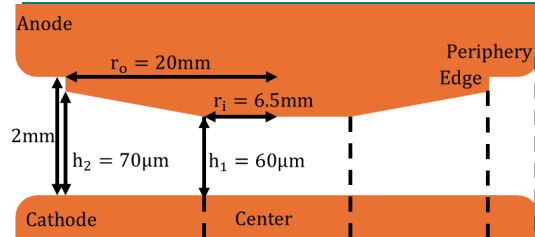
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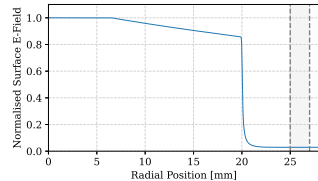


Light visual bridge: 2025 showed the model and a local cross-section; 2026 delivers the large-area structural comparison on one electrode. Do not quote the 75 percent effect here.

One cathode turns field exposure into a controlled coordinate



- Heat-treated OFE Cu, pulsed DC
- 1 μ s pulses at 1 kHz
- Peak field ~ 80 MV/m at center
- Nine EBSD regions: center, edge, periphery, and external reference



The field gradient gives us internal controls on a single specimen.

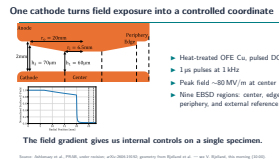
Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192; geometry from Bjelland et al. — see V. Bjelland, this morning (10:00).

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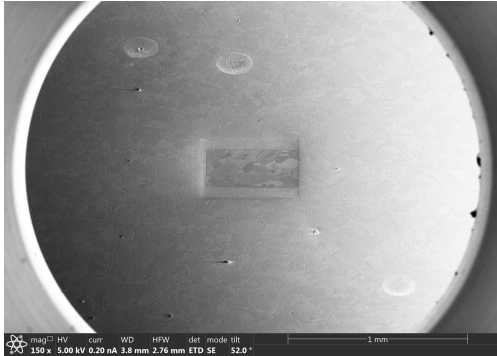
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One cathode turns field exposure into a controlled coordinate

The sloped anode maps radius onto local field: highest at the flat center, decaying across the sloped gap, and below about 2.5 MV/m beyond the anode radius. Comparing positions on one cathode removes cathode-to-cathode preparation differences, and a separate identically treated but unexposed cathode gives an external reference. That geometry is the “more comparative forensics” design from last year’s outlook.



EBSD maps plastic activity far from breakdown craters



Cleaned ROI is hundreds of microns from visible craters.

We measure the cumulative response to the field, not crater damage.

Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192; information depth after Chen et al., Ultramicroscopy 111, 1488 (2011) and

Drouin et al., Scanning 29, 92 (2007).

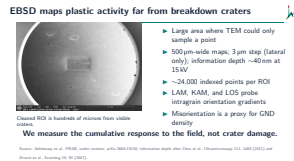
- ▶ Large area where TEM could only sample a point
- ▶ 500 μm -wide maps; 3 μm step (lateral only); information depth ~ 40 nm at 15 kV
- ▶ $\sim 24,000$ indexed points per ROI
- ▶ LAM, KAM, and LOS probe intragrain orientation gradients
- ▶ Misorientation is a proxy for GND density

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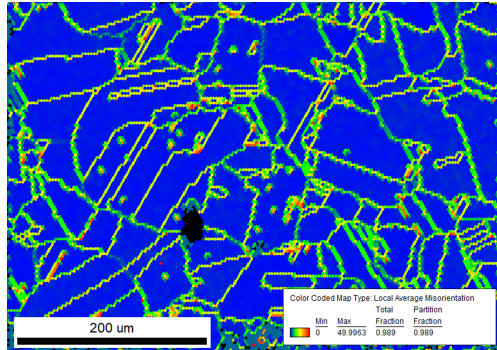
Does conditioning leave a structural memory?

└ EBSD maps plastic activity far from breakdown craters

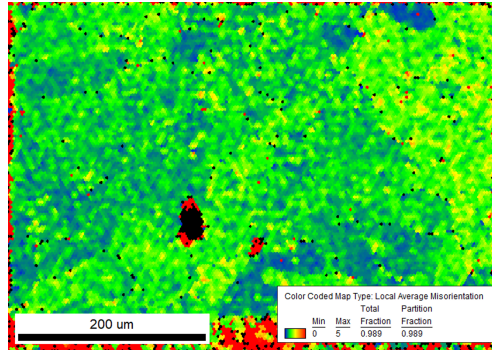
EBSD is chosen precisely because it is large-area where TEM is local. A shallow FIB clean removes oxide, under 10 nm by AFM, then we map orientation gradients. The 3 micron step and kernel fix only the lateral resolution; the backscattered electrons that carry the pattern come from the top 40 nanometers or so of copper at 15 kV, so this is a near-surface measurement. All ROIs are deliberately far from craters, so we measure the cumulative material response to the field, not arc damage. LAM is the primary metric; KAM and LOS check that the result does not depend on one analysis choice.



How to read a LAM map: wide-angle vs low-angle



0–50°: grain boundaries dominate



0–5° scale: intragrain curvature

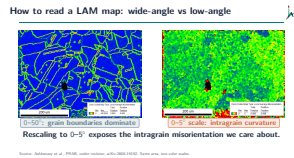
Rescaling to 0–5° exposes the intragrain misorientation we care about.

Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192. Same area, two color scales.

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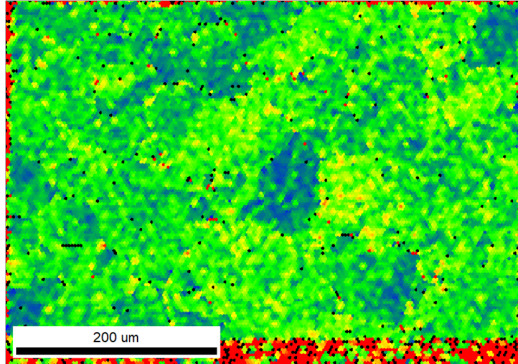
Does conditioning leave a structural memory?

└─ How to read a LAM map: wide-angle vs low-angle

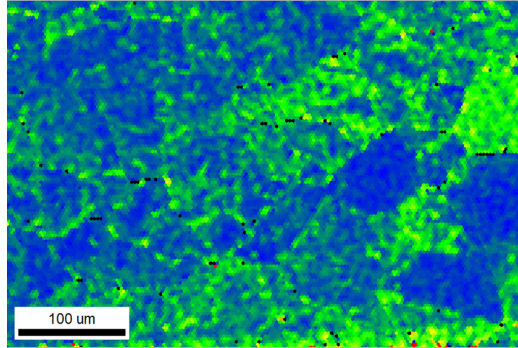


Teach the metric before the comparison. At the wide 0–50 degree scale only grain boundaries are visible. Rescaling the same map to 0–5 degrees reveals the intragrain lattice curvature—the low-angle misorientation that proxies geometrically necessary dislocations. Every LAM map that follows uses this same 0–5 degree scale.

High-field copper contains more intragrain curvature



Field-exposed center



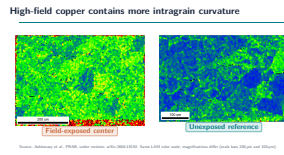
Unexposed reference

Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192. Same LAM color scale; magnifications differ (scale bars 200 μm and 100 μm).

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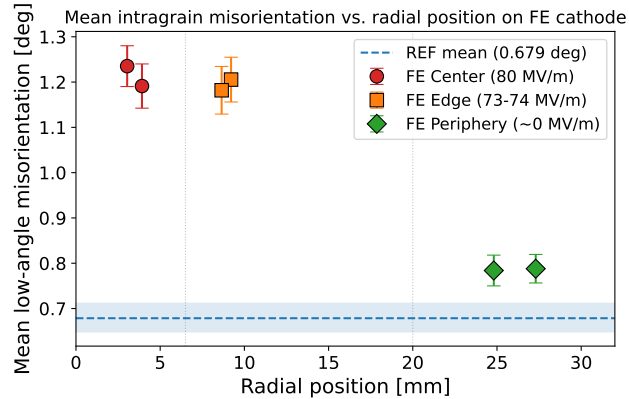
Does conditioning leave a structural memory?

└ High-field copper contains more intragrain curvature



Now the comparison. Look at the spatial texture, not just the average color: the field-exposed region carries far more high-misorientation structure inside the grains. The pair shown is representative; the quantitative conclusion uses all nine ROIs.

The signal separates into three exposure tiers



- Center and edge: $\sim 1.2^\circ$
- External reference: $\sim 0.68^\circ$
- High field is $\sim 75\%$ higher
- Grain size does not follow this hierarchy

The ordering follows field history, not sample position or grain size.

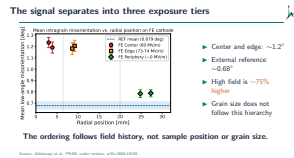
Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192.

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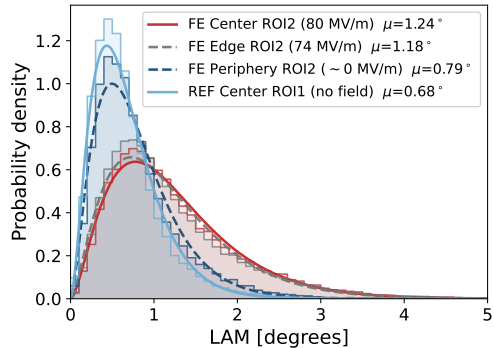
Does conditioning leave a structural memory?

The signal separates into three exposure tiers

Three tiers: high-field center and edge are highest and similar; the field-exposed periphery is intermediate; unexposed references are lowest. The periphery is powerful because it shared machining, heat treatment, mounting, vacuum, and handling with the high-field regions but saw at most about 2.5 MV/m—a low-field control, not a zero-field one. Its intermediate value is consistent with residual fringe-field exposure. The 75 percent figure characterizes this cathode and history—not a universal constant.



Field exposure builds a high-misorientation tail



LAM tail above 2°

~0.14 field-exposed
~0.016 reference
~8× increase

Gamma shape $k \approx 2.7$ fixed; the
scale roughly doubles.

The whole distribution stretches; a uniform offset cannot do that.

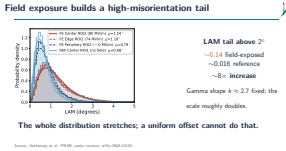
Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192.

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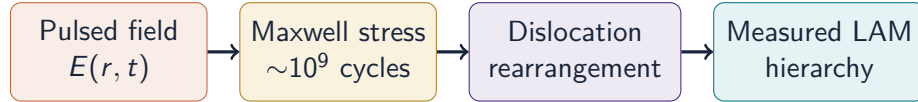
Does conditioning leave a structural memory?

Field exposure builds a high-misorientation tail

The mean is not the whole story. The low-angle distribution broadens and grows a long tail: the fraction above two degrees rises roughly eightfold. The gamma shape parameter stays near 2.7 while the scale roughly doubles—a fixed-shape stretch, the signature of progressive deformation. LAM, KAM, and LOS agree, and KS, chi-squared, and Anderson–Darling all separate the distributions; every center or edge ROI differs from the unexposed ones with a KS distance above 0.25. I quote effect sizes and grain-corrected errors rather than the extreme pixel-pair p-values.



The large-area test supports the model's prediction



- To our knowledge, the first **large-area** observation that conditioned regions carry a different near-surface dislocation structure, as MDDF predicted.
- No conflict with the 2025 denuded zone: one mechanism, two starting populations (hard Cu depletes; heat-treated Cu builds curvature).
- The three-tier ordering also tracks the conditioning variable E_S : a *candidate* structural basis for it.

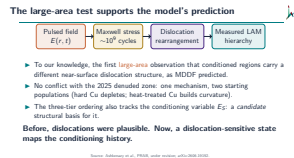
Before, dislocations were plausible. Now, a dislocation-sensitive state maps the conditioning history.

Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192.

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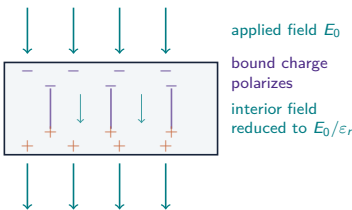


Here is the payoff. The measured hierarchy is the large-area structural signature the model predicted and the STEM saw locally. The surface denuded zone and the EBSD curvature are not in conflict: they are one mechanism acting on two very different starting populations. In dense, as-machined copper the cyclic stress lets near-surface dislocations escape to the surface or annihilate, which is the depleted layer the TEM saw; in the heat-treated copper measured here there is little to deplete, so the same load generates and rearranges dislocations, which is the curvature EBSD records. The next slide gives the length scale behind the analogy; the depth connection itself is still open. Only now, and only in one line, do I connect to E_S : the spatial ordering matches the simulated E_S profile, so the dislocation population is a candidate structural basis for it. I never say we measured E_S .

Elastic screening: dislocations respond like bound charge



Dielectric in an applied field

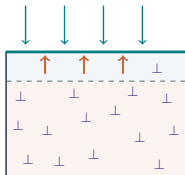


applied field E_0

bound charge polarizes

interior field reduced to E_0/ϵ_r

Copper under the Maxwell traction



field E stops at the surface charge layer

$\sigma_M = \epsilon_0 E^2 / 2$ enters as elastic stress
denuded ~ 200 nm
 $\approx 8 \ell_D$ (hard Cu)

ensemble screens internal stress
over $\ell_D = 1/(4.2\sqrt{\rho}) \approx 25$ nm

applied field $\leftrightarrow$ Maxwell traction · bound charge $\leftrightarrow$ dislocations (dipole regime) · Debye length $\leftrightarrow \ell_D$ · screened interior field $\leftrightarrow$ screened internal stress of the depleted layer

2D single-slip theory at $\rho \sim 10^{14} \text{ m}^{-2}$; no theory yet treats cyclic loading with a free surface, so the ~ 200 nm thickness is observed, not predicted.

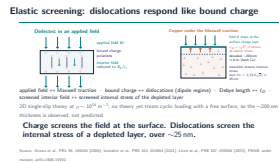
Charge screens the field at the surface. Dislocations screen the internal stress of a depleted layer, over ~ 25 nm.

Source: Groma et al., PRL 96, 165503 (2006); Lemaître et al., PRE 104, 024904 (2021); Livne et al., PRE 107, 055004 (2023); PRAB, under revision; arXiv:2606.19192.

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Does conditioning leave a structural memory?

└ Elastic screening: dislocations respond like bound charge



Picture the dislocation population as the bound charge of a dielectric: the Maxwell traction is the applied field, the dislocations rearrange in response, and in Groma's Debye-like theory the internal stress of any depleted or enriched layer is screened by the surrounding ensemble over about 25 nanometers for as-machined copper. Note the contrast on the right: the electric field itself is screened by surface charge within a nanometre and never enters the metal, whereas the load enters as elastic stress, so the electrostatic screening length says nothing about depth. The STEM denuded zone in hard copper is about eight of these screening lengths thick; no theory yet treats a cyclically loaded free surface, so that thickness is an observation, not a prediction.

Copper establishes the effect, not universality



What is established

- ▶ Large-area structural difference
- ▶ Three independent metrics
- ▶ Internal and external controls
- ▶ Supports the MDDF prediction

What remains open

- ▶ One conditioned cathode, one experiment
- ▶ Full depth profile
- ▶ Grain-orientation dependence
- ▶ More conditioned cathodes
- ▶ Cross-material generality

Call for cooperation: more cathodes, other materials, other facilities, other probes of the conditioned state (cf. M. Coman, 11:30) — let's compare notes.

A stronger test changes both the material and the loading history.

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What is established	What remains open
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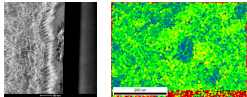
The honest limitations: one conditioned cathode and one external reference; field level and breakdown density covary; LAM is a relative proxy. Invite collaboration on replication and cross-facility comparisons. Depth matters because the STEM depletion was seen in hard copper and the EBSD curvature in heat-treated copper, each in the top few hundred nanometers at most; a depth profile on one material would show whether they are two ends of one redistribution. The next experiments therefore add depth, and change the material and loading mode.

Next: go deeper, and change the material



Depth

- ▶ Matched high-field and reference FIB lamellae; TEM/STEM + EBSD through the top $\sim 1\mu\text{m}$
- ▶ Artifact controls from the 2025 “red herrings”
- ▶ One depth profile from STEM depletion to EBSD curvature?



Depth continues the Uppsala STEM line; material tests the mechanism itself.

Material & loading

- ▶ Stainless steel: lower stacking-fault energy, different dislocation mobility and slip
- ▶ Continuous DC instead of pulsed loading
- ▶ Possible metastable phase transformation



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Depth continues the Uppsala STEM line; material tests the mechanism itself.

Source: Interim work plan (2026); RFX project plan.

Two orthogonal directions. Depth is a direct continuation of the STEM work: pilot lamellae from a mapped high-field ROI and a matched reference, with the artifact controls we learned we need in 2025. The material axis is the RFX stainless-steel collaboration, which changes stacking-fault energy, dislocation mobility, and the loading mode from pulsed to continuous DC.

Stainless steel: a test chosen to be hard on the model



Copper
FCC, higher SFE
pulsed DC, $\sim 10^9$ cycles

AISI 304L SS
metastable austenite, lower SFE
continuous DC, possible $\gamma \rightarrow \alpha'$

Design: apex-to-periphery gradient; two exposed electrodes, two unexposed controls.

SS tracks field cross-material structural state

SS differs material-specific carriers

no resolved change informative only with enough maps

Status: 90 h and 73 h DC, peak 59–61 MV/m, zero breakdowns (stopped at emission switch-on); all 26 EBSD maps in hand; the exposed apex sits inside the unexposed range on every quantity tested, so no field indication yet.



Consorzio RFX — De Lorenzi, Pilan, Spada

Any outcome constrains the mechanism; the test is not pass-or-fail.

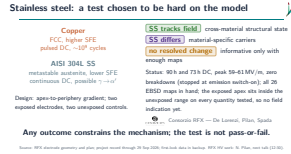
Source: RFX electrode geometry and plan; project record through 29 Sep 2026; first-look data in backup. RFX HV work: N. Pilan, next talk (12:30).

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└ Stainless steel: a test chosen to be hard on the model

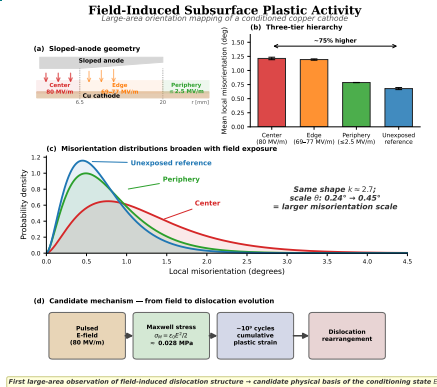
Stainless steel is chosen because it is hard: it changes the stacking-fault energy, the dislocation mobility, the loading mode, and adds a possible austenite-to-martensite transformation. That is what makes it discriminating. A field-correlated response supports a broad structural mechanism; a qualitatively different response identifies which material degrees of freedom store the state; a null result is informative only once the statistics can bound the sensitivity, which at the current two to four maps per group they cannot; with enough maps it may separate cyclic from continuous loading. Where we stand: the two exposed electrodes ran about 90 and 73 hours of continuous DC at up to 61 megavolts per meter with no full breakdowns. All 26 EBSD maps are in hand, and the exposed apex sits inside the unexposed range on every quantity we tested, so there is no field indication yet. The runs stopped at emission switch-on with zero breakdowns. One hypothesis we will test: as-manufactured 304L is already in its conditioned dislocation state, so the run never engaged conditioning. An early apex-versus-side contrast turned out to reproduce on a never-installed reference, so it is forming geography. No stainless-steel field effect is claimed. First-look SEM and EBSD are in backup.



Conditioning appears to leave a subsurface memory



1. Our model predicts conditioning evolves the near-surface dislocation structure.
2. 2025 STEM saw it locally; **large-area EBSD now supports it** (~75% higher misorientation).
3. Depth and stainless steel test whether the memory survives a change of scale, alloy, and loading.



Last year: “other materials? more samples?” — here is the large-area answer.

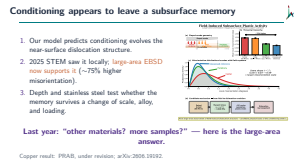
Copper result: PRAB, under revision; arXiv:2606.19192.

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Does conditioning leave a structural memory?

Conditioning appears to leave a subsurface memory

Close the loop from the title and from last year. The model predicted a structural memory; the STEM saw it at one spot; the large-area EBSD supports it across the electrode with a 75 percent effect and a three-tier hierarchy. The stainless-steel and depth programs are the next round of comparative forensics I promised in 2025. Invite collaboration on more cathodes and cross-material tests. Cite the manuscript as under revision at PRAB, arXiv:2606.19192.

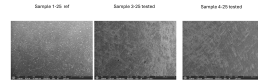


Backup: RFX first look (preliminary)

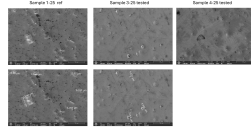
Preliminary — no field effect claimed

- ▶ AISI 304L at RFX HVPTF; continuous DC: 90 h (#3-25), 73 h (#4-25); peak 59–61 MV/m; zero breakdowns (stopped at emission switch-on)
- ▶ SEM: “exploded” near-surface inclusions (3–7 μm) on field-exposed samples; reference cavities round and clean
- ▶ EBSD: #1-25, #2-25, #3-25 index (97–99.5%); #4-25: two maps, one an outlier on confidence index, position unrecorded
- ▶ Apex-vs-side contrast reproduces on never-installed #2-25: forming geography

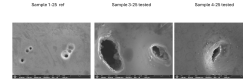
Source: RFX project record through 29 Sep 2026; SEM report, Inna Popov (20 May 2026).



Ref vs field-exposed overview



Sectioned electrode

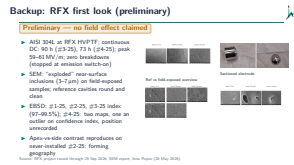


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Does conditioning leave a structural memory?

└ Backup: RFX first look (preliminary)

Backup for questions on the stainless-steel first look. SEM survey plus EBSD quality: three samples index well; the field-exposed #4-25 has two maps, one of them an outlier on confidence index, and its position on the electrode was not recorded, so the cause is open. The July apex-versus-side contrast reproduced on a never-installed reference, so it reflects forming geography. No structural field effect is claimed.



Backup: RFX sample and data status (29 Sep 2026)



Sample	Role	Exposure	Status
#1-25	reference	none	SEM; EBSD apex + side, 97% indexed
#2-25	reference	none	EBSD apex (18 Aug, 3 maps) + side; LAM done
#3-25	field-exposed	90 h DC, peak 61 MV/m	SEM; EBSD apex + side, 99.5% indexed; LAM done
#4-25	field-exposed	73 h DC, peak 59 MV/m	SEM/EDS; EBSD two maps, one an outlier on confidence index

- ▶ Received since July: conditioning report and $E(\theta)$; alloy confirmed as 304L, mill certificate unrecoverable; electropolishing partly documented; all 26 catalogue EBSD maps (36 scans), 28 Sep.
- ▶ XRD: δ -ferrite content differs between the reference and exposed pairs, so the clean comparison is apex vs side within one electrode.
- ▶ Wide-vs-narrow map class = grain size and indexing quality (also on never-installed #2-25), not field.
- ▶ Still open: thermal history of #3/#4 (vacuum firing?); matrix Cr/Ni EDS on all four under identical conditions; SEM θ -survey as the first exposure marker.

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Does conditioning leave a structural memory?

└ Backup: RFX sample and data status (29 Sep 2026)

Use only if asked about readiness or missing inputs. Reflects the written project record through 29 September 2026.

Sample	Role	Exposure	Status
#1-25	reference	none	SEM; EBSD apex + side, 97% indexed
#2-25	reference	none	EBSD apex (18 Aug, 3 maps) + side; LAM done
#3-25	field-exposed	90 h DC, peak 61 MV/m	SEM; EBSD apex + side, 99.5% indexed; LAM done
#4-25	field-exposed	73 h DC, peak 59 MV/m	SEM/EDS; EBSD two maps, one an outlier on confidence index

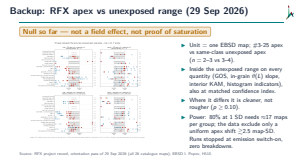
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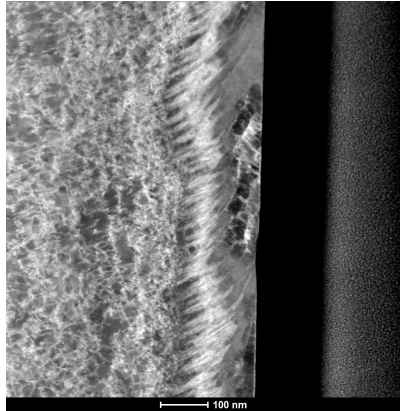
└ Backup: RFX apex vs unexposed range (29 Sep 2026)



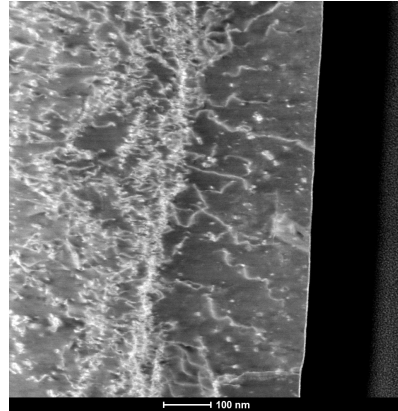
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Backup: field-exposed vs reference STEM (2025)



Field-exposed, 300 K: fewer walls



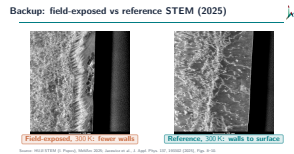
Reference, 300 K: walls to surface

Source: HUJI STEM (I. Popov), MeVArc 2025; Jacewicz et al., J. Appl. Phys. 137, 193302 (2025), Figs. 8–10.

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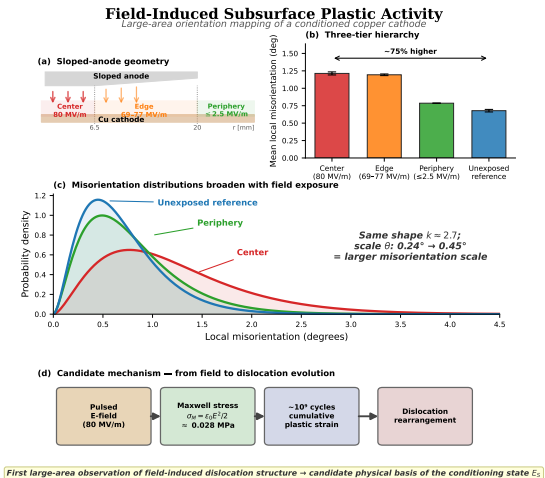
Does conditioning leave a structural memory?

└ Backup: field-exposed vs reference STEM (2025)



The field-exposed versus reference STEM pair from the 300 K cathode. In the published paper this pair shows a strong reduction of dislocation walls (Fig. 9); the zone the paper calls dislocation-denuded and clearly visible is the 30 K cathode (Fig. 10). No thickness is stated in the paper.

Backup: copper result in one figure



2026-10-02

Does conditioning leave a structural memory?

Backup: copper result in one figure

Compact answer to questions about geometry, distributions, or the proposed mechanism.

